

In right angled $\triangle ABC$,

$$\tan 45^\circ = \frac{BC}{AB} = \frac{b}{x} \Rightarrow x = b \quad \dots(i)$$

and in right angled $\triangle ABD$,

$$\tan 60^\circ = \frac{DB}{BA} = \frac{DC + CB}{AB} = \frac{100 + b}{x}$$

$$\Rightarrow \sqrt{3} = \frac{100 + b}{b} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow \sqrt{3}b = 100 + b$$

$$\Rightarrow (\sqrt{3} - 1)b = 100 \Rightarrow b = \frac{100}{\sqrt{3} - 1}$$

$$\Rightarrow b = \frac{100}{(\sqrt{3} - 1)} \times \frac{(\sqrt{3} + 1)}{(\sqrt{3} + 1)}$$

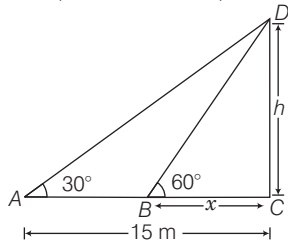
$$= 50(\sqrt{3} + 1) \text{ m}$$

Hence, the height of the building is $50(\sqrt{3} + 1)$ m.

5. (c) Let the height of the tower be b m and length of the shadow (BC) be x m. In right angled $\triangle ACD$,

$$\tan 30^\circ = \frac{CD}{AC}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{b}{15} \Rightarrow b = \frac{15}{\sqrt{3}} \text{ m} \quad \dots(i)$$



and in right angled $\triangle BCD$,

$$\tan 60^\circ = \frac{CD}{BC} \Rightarrow \sqrt{3} = \frac{b}{x} \Rightarrow \frac{b}{\sqrt{3}} = x$$

$$\therefore x = \frac{15}{3} = 5 \text{ m}$$

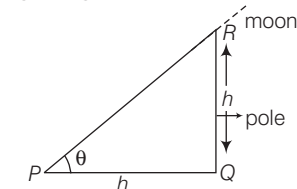
[from Eq. (i)]

Hence, the length of shadow is 5 m.

When sun's altitude is 60° .

6. (b) Let the height of pole be b m, then $PQ = RQ = b$

In right angled $\triangle PRQ$,



$$\tan \theta = \frac{RQ}{PQ} = \frac{b}{h} = 1$$

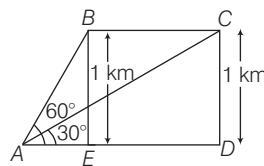
$$\Rightarrow \tan \theta = 1 = \tan 45^\circ \therefore \theta = 45^\circ$$

Hence, the angle of elevation of moon is 45° .

7. (b) Let A be the observer. When the aeroplane was at point B the angle of elevation was 60° and after 10 s when it was at point C the angle of elevation was 30° .

Here, in the figure,

$$BE = CD = 1 \text{ km} = 1000 \text{ m}$$



In right angled $\triangle BAE$,

$$\tan 60^\circ = \frac{BE}{AE} = \frac{1000}{AE}$$

$$\Rightarrow AE = 1000 \cot 60^\circ$$

$$= \frac{1000}{\sqrt{3}} \text{ m} \quad \dots(i)$$

In right angled $\triangle ACD$,

$$\tan 30^\circ = \frac{DC}{AD} = \frac{1000}{AD}$$

$$\Rightarrow AD = 1000 \cot 30^\circ = 1000 \sqrt{3}$$

$$\therefore ED = AD - AE = 1000 \sqrt{3} - \frac{1000}{\sqrt{3}} = \frac{2000}{\sqrt{3}} \text{ m}$$

So, distance travelled by plane in 10 s, then

$$BC = DE = \frac{2000}{\sqrt{3}} \text{ m}$$

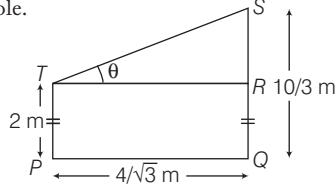
$$\therefore \text{Speed of aeroplane} = \frac{\text{Distance}}{\text{Time}}$$

$$= \frac{\frac{2000}{\sqrt{3}}}{10} \text{ m/s} = \frac{2000 \times 60 \times 60}{10 \sqrt{3} \times 1000} \text{ km/h}$$

$$= 240 \sqrt{3} \text{ km/h}$$

Hence, the uniform speed of the aeroplane is $240\sqrt{3}$ km/h.

8. (d) Let TP be the man and SQ be the pole.



Let $\angle STR = \theta$

Now, $SR = SQ - RQ$

$$= \frac{10}{3} - 2 = \frac{4}{3} \text{ m} [\because RQ = TP]$$

In right angled $\triangle STR$,

$$\tan \theta = \frac{SR}{TR} = \frac{SR}{PQ} \quad [\because TR = PQ]$$

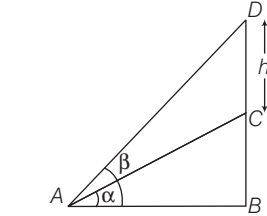
$$= \frac{4}{3} \times \frac{\sqrt{3}}{4} = \frac{1}{\sqrt{3}} = \tan 30^\circ$$

$$\therefore \theta = 30^\circ$$

9. (d) Let BC be the tower and CD be the flagstaff.

$$\therefore \angle BAC = \alpha \text{ and } \angle BAD = \beta$$

In right angled $\triangle ABC$,



$$\tan \alpha = \frac{BC}{AB} \quad \dots(i)$$

and in right angled $\triangle ABD$,

$$\frac{BD}{AB} = \frac{BC + b}{AB} = \tan \beta \quad \dots(ii)$$

On dividing Eq. (ii) by Eq. (i), we get

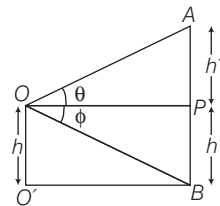
$$\frac{BC + b}{BC} = \frac{\tan \beta}{\tan \alpha}$$

$$\Rightarrow (BC + b) \tan \alpha = BC \tan \beta$$

$$\Rightarrow BC (\tan \beta - \tan \alpha) = b \tan \alpha$$

$$\therefore BC = \frac{b \tan \alpha}{\tan \beta - \tan \alpha}$$

10. (b) Let O be the window and AB be the house on the opposite side of the street.



Let $AP = b'$

and $BP = OO' = b$

In right angled $\triangle AOP$,

$$\tan \theta = \frac{AP}{OP} = \frac{b'}{OP} \quad \dots(i)$$

and in right angled $\triangle BOP$,

$$\tan \phi = \frac{BP}{OP} = \frac{b}{OP} \quad \dots(ii)$$

$$\therefore \frac{b'}{b} = \frac{\tan \theta}{\tan \phi} \quad [\text{by Eq. (i) } \div \text{ Eq. (ii)}]$$

$$\Rightarrow b' = b \tan \theta \cot \phi$$

$\therefore$ Height of the house

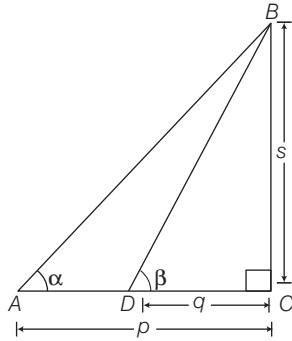
$$AB = AP + PB = b + b'$$

$$= b \tan \theta \cot \phi + b$$

$$= b [\tan \theta \cot \phi + 1]$$

11. (c) In right angled $\triangle BAC$,

$$\tan \alpha = \frac{BC}{AC} = \frac{s}{p} \Rightarrow p = \frac{s}{\tan \alpha} \quad \dots(i)$$



In right angled $\triangle BDC$,

$$\tan \beta = \frac{BC}{DC} = \frac{s}{q} \Rightarrow q = \frac{s}{\tan 3\alpha} \quad \dots(ii)$$

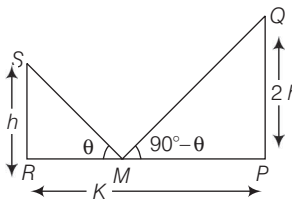
[$\because \beta = 3\alpha$, given]

On subtracting Eq. (ii) from Eq. (i), we get

$$p - q = \frac{s}{\tan \alpha} - \frac{s}{\tan 3\alpha}$$

$$= s \left(\frac{\tan 3\alpha - \tan \alpha}{\tan \alpha \tan 3\alpha} \right)$$

12. (b) Let PQ and RS be the two posts such that $PQ = 2RS$ [given]



Given, M is the mid-point of RP .

$$\therefore RM = PM = \frac{K}{2} \quad [\because RP = K \text{ given}]$$

Let $\angle RMS = \theta$

$$\therefore \angle QMP = 90^\circ - \theta$$

Let $RS = h$, then $PQ = 2h$

Now, in right angled $\triangle PMQ$,

$$\tan (90^\circ - \theta) = \frac{PQ}{MP}$$

$$\Rightarrow \cot \theta = \frac{2h}{K/2} = \frac{4h}{K}$$

$$\text{or } \cot \theta = \frac{4h}{K} \quad \dots(i)$$

In right angled $\triangle SRM$,

$$\tan \theta = \frac{SR}{RM} = \frac{h}{K/2}$$

$$\Rightarrow \frac{2h}{K} = \tan \theta \quad \dots(ii)$$

On multiplying Eq. (i) by Eq. (ii), we get

$$\Rightarrow \frac{4h}{K} \times \frac{2h}{K} = 1$$

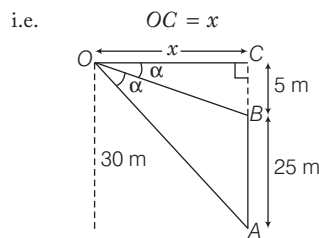
$$\Rightarrow 8h^2 = K^2 \Rightarrow h^2 = \frac{K^2}{8}$$

$$\therefore h = \frac{K}{2\sqrt{2}} \text{ m}$$

Hence, the height of shorter post is

$$\frac{K}{2\sqrt{2}} \text{ m.}$$

13. (d) Let O be an observer at a height 30 m. Let x be the distance of observer from the top of the flagstaff.



Here, AB be the tower of the flag staff = 25 m and BC be the flag staff = 5 m

In right angled $\triangle OBC$,

$$\tan \alpha = \frac{BC}{OC} = \frac{5}{x}$$

$$\Rightarrow x = 5 \cot \alpha \quad \dots(i)$$

In right angled $\triangle OCA$,

$$\tan 2\alpha = \frac{AC}{OC} = \frac{30}{x} \quad [\because AC = AB + BC]$$

$$\therefore \tan 2\alpha = \frac{30}{5 \cot \alpha} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow \tan 2\alpha = 6 \tan \alpha$$

$$\Rightarrow \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = 6 \tan \alpha$$

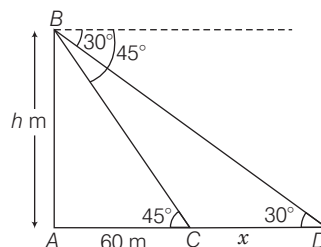
$$\left[\because \tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha} \right]$$

$$\Rightarrow 3 - 3 \tan^2 \alpha = 1 \Rightarrow \tan^2 \alpha = \frac{2}{3}$$

$$\Rightarrow \tan \alpha = \sqrt{\frac{2}{3}} \text{ or } \cot \alpha = \sqrt{3/2}$$

$$\text{or } x = 5 \sqrt{\frac{3}{2}} \text{ m } [\text{from Eq. (i)}]$$

14. (a) Let height of tower AB be h m and distance between C and D be x m.



In right angled $\triangle ACB$, $\tan 45^\circ = \frac{AB}{AC}$

$$\Rightarrow 1 = \frac{h}{60} \Rightarrow h = 60 \text{ m} \quad \dots(i)$$

Now, in right angled $\triangle ADB$,

$$\tan 30^\circ = \frac{AB}{AD} = \frac{AB}{AC + CD} = \frac{60}{60 + x}$$

[from Eq. (i)]

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{60}{60 + x} \Rightarrow 60 + x = 60\sqrt{3}$$

$$\Rightarrow x = 60(\sqrt{3} - 1) = 60(1.73 - 1)$$

$$= 60 \times 0.73 = 43.8 \text{ m}$$

Now, given time = 5 s

We know that, Speed = $\frac{\text{Distance}}{\text{Time}}$

$$\therefore \text{Speed of boat} = \frac{43.8}{5} \times \frac{18}{5} = \frac{788.4}{25}$$

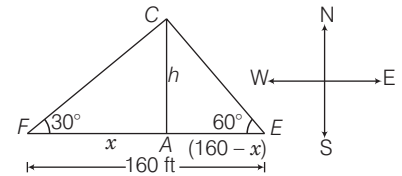
$$= 31.5 \text{ km/h}$$

15. (a) Let $AC = h$ ft = Height of a tree and x ft = Distance between A and F then $AE = (160 - x)$ ft

In right angled $\triangle AFC$,

$$\tan 30^\circ = \frac{AC}{AF} = \frac{h}{x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x}$$

$$\Rightarrow x = \sqrt{3}h \quad \dots(i)$$



and in right angled $\triangle AEC$,

$$\tan 60^\circ = \frac{AC}{AE} = \frac{h}{160 - x}$$

$$\Rightarrow \sqrt{3} = \frac{h}{160 - x} \quad [\because \tan 60^\circ = \sqrt{3}]$$

$$\Rightarrow \sqrt{3}(160 - x) = h$$

$$\Rightarrow \sqrt{3}(160 - \sqrt{3}h) = h \quad [\text{from Eq. (i)}]$$

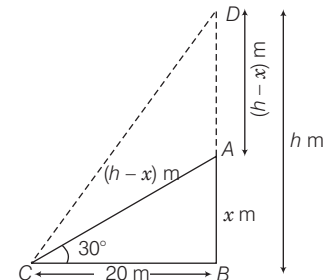
$$\Rightarrow 160\sqrt{3} - 3h = h \Rightarrow 4h = 160\sqrt{3}$$

$$\therefore h = 40\sqrt{3} \text{ ft}$$

Hence, the height of the tree is $40\sqrt{3}$ ft.

16. (b) Let the height of the post be h m, and $AB = x$ m.

Given, $BC = 20$ m



In right angled $\triangle ABC$,

$$\tan 30^\circ = \frac{AB}{BC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{x}{20}$$

$$\Rightarrow x = \frac{20}{\sqrt{3}} \text{ m} \quad \dots(i)$$

$$\text{and } \cos 30^\circ = \frac{BC}{AC} = \frac{20}{b-x}$$

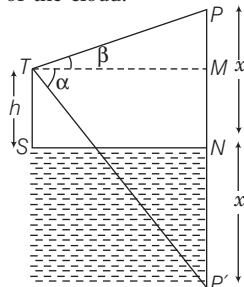
$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{20}{b-x} \Rightarrow b-x = \frac{40}{\sqrt{3}}$$

From Eq. (i), putting the value of x ,

$$b = \frac{40}{\sqrt{3}} + \frac{20}{\sqrt{3}} = \frac{60}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 20\sqrt{3} \text{ m}$$

Hence, the height of the post is $20\sqrt{3}$ m.

17. (c) Let P be the cloud and P' its image in the lake. Let T be the point 'b' m above the surface of the lake and let x be the height of the cloud.



$$\therefore ST = b$$

$$\text{Also, } NP = NP' = x \quad [\text{say}]$$

$$\text{Then, } PM = x - b$$

$$P'M = x + b$$

In right angled $\triangle PTM$,

$$\frac{PM}{TM} = \tan \beta$$

$$\therefore x - b = TM \tan \beta \quad \dots(i)$$

In right angled $\triangle P'TM$,

$$\frac{P'M}{TM} = \tan \alpha$$

$$x + b = TM \tan \alpha \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{x-b}{x+b} = \frac{\tan \beta}{\tan \alpha}$$

Using componendo and dividendo rule, we get

$$\frac{x-b+x+b}{x-b-x-b} = \frac{\tan \beta + \tan \alpha}{\tan \beta - \tan \alpha}$$

$$\Rightarrow \frac{2x}{-2b} = \frac{\tan \beta + \tan \alpha}{\tan \beta - \tan \alpha}$$

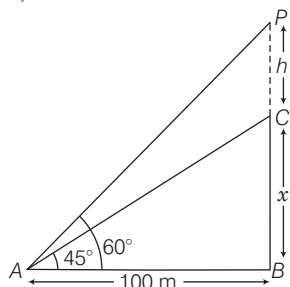
$$\Rightarrow \frac{x}{b} = \frac{\tan \beta + \tan \alpha}{\tan \alpha - \tan \beta}$$

$$\begin{aligned} \Rightarrow \frac{x}{b} &= \frac{\sin \beta \cos \alpha + \cos \beta \sin \alpha}{\sin \alpha \cos \beta - \cos \alpha \sin \beta} \\ &= \frac{\sin(\alpha + \beta)}{\sin(\alpha - \beta)} \end{aligned}$$

$$\therefore x = b \sin(\alpha + \beta) \operatorname{cosec}(\alpha - \beta)$$

18. (c) Let the height of the incomplete pillar be x m and the increasing height be $PC = h$ m.

Given, $AB = 100$ m



In right angled $\triangle ABC$,

$$\tan 45^\circ = \frac{BC}{AB} = \frac{x}{100} \Rightarrow x = 100 \text{ m} \dots(i)$$

and in right angled $\triangle APB$,

$$\tan 60^\circ = \frac{PB}{AB} = \frac{PC + BC}{AB} = \frac{x+h}{100}$$

$$\Rightarrow x + h = 100\sqrt{3}$$

$$\Rightarrow h = 100\sqrt{3} - x = 100\sqrt{3} - 100$$

[from Eq. (i)]

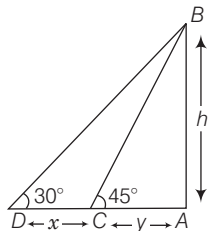
$$\therefore h = 100(\sqrt{3} - 1) \text{ m}$$

Hence, the height of the incomplete pillar is $100(\sqrt{3} - 1)$ m.

19. (c) Let AB be the tower and C, D be the two positions of the car.

Then, $\angle ACB = 45^\circ$, $\angle ADB = 30^\circ$.

Let $AB = b$, $CD = x$ and $AC = y$.



In right angled $\triangle ABC$, we get

$$\frac{AB}{AC} = \tan 45^\circ = 1 \Rightarrow \frac{b}{y} = 1$$

$$\Rightarrow y = b \quad \dots(i)$$

In right angled $\triangle ABD$, we get

$$\frac{AB}{AD} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{b}{x+y} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow x + y = \sqrt{3}b \quad \dots(ii)$$

$$\therefore x = \sqrt{3}b - b = b(\sqrt{3} - 1)$$

[by Eqs. (i) and (ii)]

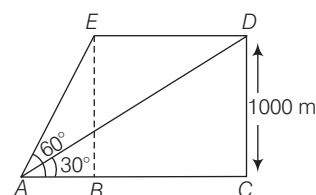
Now, $b(\sqrt{3} - 1)$ is covered in 12 min.

So, b will be covered in

$$\left[\frac{12}{b(\sqrt{3} - 1)} \times b \right] = \frac{12}{(\sqrt{3} - 1)} \text{ min.}$$

$$= \left(\frac{1200}{73} \right) \text{ min} \approx 16 \text{ min } 23 \text{ s}$$

20. (b) Let the radar base is at point A . The plane is at point D in the first sweep and at point E in the second sweep. The distance it covers in the one minute interval is DE .



From the figure,

In right angled $\triangle ADC$, we get

$$\tan 30^\circ = \frac{DC}{AC} = \frac{1000}{AC} \Rightarrow AC = \frac{1000}{\tan 30^\circ}$$

Similarly, in $\triangle EAB$, we get

$$\tan 60^\circ = \frac{EB}{AB} = \frac{1000}{AB} \Rightarrow AB = \frac{1000}{\tan 60^\circ}$$

Total distance covered by plane in 1 min, then

$$DE = AC - AB$$

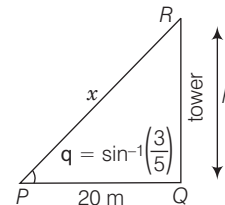
$$\begin{aligned} DE &= \frac{1000}{\tan 30^\circ} - \frac{1000}{\tan 60^\circ} \\ &= 1000\sqrt{3} - \frac{1000}{\sqrt{3}} = 1154.70 \text{ m} \end{aligned}$$

The speed of the plane is given by

$$s = \text{distance covered/time taken}$$

$$= DE / 60 = 19.25 \text{ m/s.}$$

21. (c)



Let P be the point of observation and QR be the tower.

Given that, $\theta = \sin^{-1}\left(\frac{3}{5}\right)$, $PQ = 20$ m.

Let the height of the tower, $QR = b$ and $PR = x$

From the right angled $\triangle PQR$,

$$\sin \theta = \frac{QR}{PR} \Rightarrow \sin \left[\sin^{-1}\left(\frac{3}{5}\right) \right] = \frac{b}{x}$$

$$\Rightarrow \frac{3}{5} = \frac{b}{x} \Rightarrow x = \frac{5b}{3} \quad \dots(i)$$

From pythagoras theorem, we get,

$$PQ^2 + QR^2 = PR^2$$

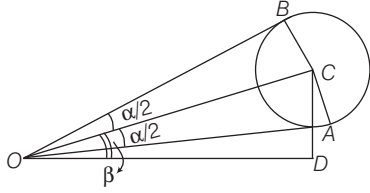
$$\Rightarrow 20^2 + b^2 = x^2$$

$$\Rightarrow 20^2 + b^2 = \left(\frac{5b}{3}\right)^2 \quad [\text{using Eq. (i)}]$$

$$\Rightarrow 20^2 + b^2 = \frac{25b^2}{9} \Rightarrow \frac{16b^2}{9} = 20^2$$

$$\therefore b = \frac{3 \times 20}{4} = 15 \text{ m}$$

22. (a) Let O be the position of the man's eye and C be the centre of the balloon.



$$\text{Now, } \angle BOC = \angle COA = \frac{\alpha}{2}$$

$$\text{Given, } CA = BC = r$$

$$\text{In right angled } \triangle COD, \sin \beta = \frac{CD}{OC}$$

$$\therefore CD = OC \sin \beta \quad \dots(i)$$

$$\text{In right angled } \triangle COA,$$

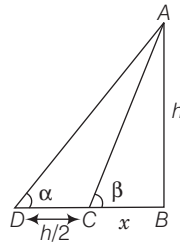
$$\sin \frac{\alpha}{2} = \frac{CA}{OC}$$

$$\therefore OC = \frac{r}{\sin \frac{\alpha}{2}} = r \operatorname{cosec} \frac{\alpha}{2}$$

$$\text{From Eq. (i), } CD = r \operatorname{cosec} \frac{\alpha}{2} \sin \beta$$

Hence, the height of centre of the balloon is $r \sin \beta \operatorname{cosec} \frac{\alpha}{2}$.

23. (a) Let AB be the tower of height h m, BC be x m and DC be $h/2$ m.



In right angled $\triangle ABD$,

$$\tan \alpha = \frac{h}{x + h/2} \Rightarrow \cot \alpha = \frac{x + h/2}{h} \quad \dots(i)$$

Now, in right angled $\triangle ABC$,

$$\tan \beta = \frac{h}{x} \Rightarrow \cot \beta = \frac{x}{h} \quad \dots(ii)$$

Subtracting eq. (ii) from eq. (i), we get

$$\cot \alpha - \cot \beta = \frac{x + h/2}{h} - \frac{x}{h}$$

$$= \frac{x + h/2 - x}{h} = \frac{h/2}{h} = \frac{1}{2}$$

24. (a) We have, $\cot \alpha - \cot \beta = 1/2$
 $\Rightarrow \cot 30^\circ - \cot \beta = 1/2$
 $\Rightarrow \cot \beta = \cot 30^\circ - 1/2$
 $= \sqrt{3} - 1/2$

25. (b) $\tan \beta = 4 \Rightarrow \cot \beta = 1/4$

we have, $\cot \alpha - \cot \beta = 1/2$

$$\Rightarrow \cot \alpha = \cot \beta + 1/2 = 1/4 + 1/2 = \frac{3}{4}$$

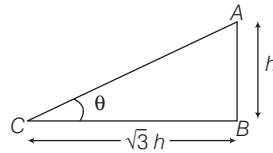
$$\text{In } \triangle ABD, \cot \alpha = \frac{x + h/2}{h}$$

$$\therefore \frac{3}{4} = \frac{x + h/2}{h}$$

$$\Rightarrow 3h = 4x + 2h \Rightarrow x = \frac{h}{4}$$

Hence, required distance is $\frac{h}{4}$

26. (a) Let θ be the angle of elevation of Sun and height of the pole be h m.



In right angled $\triangle ABC$,

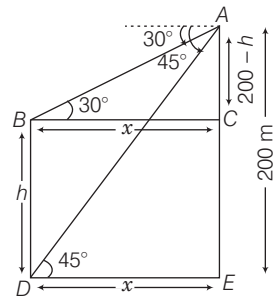
$$\tan \theta = \frac{AB}{BC} \Rightarrow \tan \theta = \frac{h}{\sqrt{3}h}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3}} = \tan 30^\circ$$

$$\therefore \theta = 30^\circ$$

Hence, the angle of elevation of Sun is 30° .

27. (d) Let AE be the height of the cliff and h m (BD) be the height of the tower.



In right angled $\triangle ABC$,

$$\tan 30^\circ = \frac{AC}{BC} = \frac{200 - h}{x}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{200 - h}{x}$$

$$\Rightarrow x = (200 - h) \sqrt{3} \quad \dots(i)$$

and in right angled $\triangle ADE$,

$$\tan 45^\circ = \frac{AE}{DE} = \frac{200}{x}$$

$$\Rightarrow 1 = \frac{200}{x} \Rightarrow x = 200 \text{ m}$$

$$\text{From Eq. (i), } 200 = (200 - h) \sqrt{3}$$

$$\Rightarrow 200 = 200\sqrt{3} - h\sqrt{3}$$

$$\Rightarrow h\sqrt{3} = 200(\sqrt{3} - 1)$$

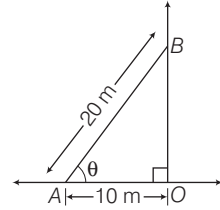
$$\therefore h = 200 \left(\frac{\sqrt{3} - 1}{\sqrt{3}} \right) \text{ m}$$

Hence, the height of tower is

$$\frac{200(\sqrt{3} - 1)}{\sqrt{3}} \text{ m.}$$

28. (c) Refer to example 3.

29. (c) Let θ be the inclination of the ladder to the horizontal.



Now, in right angled $\triangle AOB$,

$$\cos \theta = \frac{AO}{AB} = \frac{10}{20} = \frac{1}{2}$$

$$\Rightarrow \cos \theta = \cos 60^\circ$$

$$\therefore \theta = 60^\circ$$

Hence, the angle of inclination of the ladder is 60° .

Solutions (Q. Nos. 30-33)

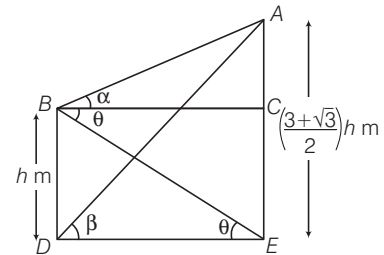
30. (b) Given, $\beta = 30^\circ$

In right angled $\triangle ADE$, $\tan \beta = \frac{AE}{DE}$

$$\tan 30^\circ = \frac{AE}{DE} \Rightarrow \frac{1}{\sqrt{3}} = \frac{AE}{DE}$$

$$\Rightarrow DE = \sqrt{3} AE = \sqrt{3} \left(\frac{3 + \sqrt{3}}{2} \right) h$$

$$\Rightarrow BC = DE = \frac{3}{2} (1 + \sqrt{3}) h \quad \dots(i)$$



$$[\because BC = DE]$$

Now, in right angled $\triangle ABC$,

$$\Rightarrow \tan \alpha = \frac{AC}{BC}$$

$$\Rightarrow BC \tan \alpha = (AE - CE)$$

$$= (AE - BD) [\because BD = CE]$$

$$\Rightarrow BC \tan \alpha = \left(\frac{3 + \sqrt{3}}{2} \right) h - h$$

$$= h \left(\frac{3 + \sqrt{3} - 2}{2} \right)$$

$$\Rightarrow \frac{3}{2} (1 + \sqrt{3}) h \tan \alpha = \left(\frac{1 + \sqrt{3}}{2} \right) h$$

[from Eq. (i)]

$$\therefore \tan \alpha = \frac{1}{3}$$

31. (a) Given, $\alpha = 30^\circ$

In right angled $\triangle ABC$,

$$\tan \alpha = \tan 30^\circ = \frac{AC}{BC}, \frac{1}{\sqrt{3}} = \frac{AC}{BC}$$

$$\begin{aligned} \Rightarrow BC &= \sqrt{3} AC = \sqrt{3} (AE - CE) \\ &= \sqrt{3} (AE - BD) [\because BD = CE] \\ &= \sqrt{3} \left(\frac{3 + \sqrt{3}}{2} - 1 \right) h \\ &= \frac{\sqrt{3}}{2} (1 + \sqrt{3}) h \quad \dots(ii) \end{aligned}$$

Now, in right angled $\triangle ADE$,

$$\begin{aligned} \tan \beta &= \frac{AE}{DE} \\ \Rightarrow \tan \beta &= \frac{AE}{BC} [\because DE = BC] \\ &= \frac{\left(\frac{3 + \sqrt{3}}{2} \right) h}{\frac{\sqrt{3}}{2} (1 + \sqrt{3}) h} = \frac{\sqrt{3}(1 + \sqrt{3})}{\sqrt{3}(1 + \sqrt{3})} \\ \therefore \tan \beta &= 1 \end{aligned}$$

32. (c) Given, $\alpha = 30^\circ$ and $h = 30$ m

In right angled $\triangle ABC$,

$$\begin{aligned} \tan \alpha &= \tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{AC}{BC} \\ \Rightarrow \frac{BC}{\sqrt{3}} &= (AE - CE) = (AE - BD) \\ \Rightarrow BC &= \sqrt{3} \left(\frac{3 + \sqrt{3}}{2} - 1 \right) h \\ \Rightarrow BC &= \sqrt{3} \left(\frac{1 + \sqrt{3}}{2} \right) \cdot 30 \\ &= (\sqrt{3} + 3) \cdot 15 \\ \therefore DE = BC &= (45 + 15\sqrt{3}) \text{ m} \\ &[\because DE = BC] \end{aligned}$$

33. (a) Given, $\beta = 30^\circ$

In right angled $\triangle ADE$, $\tan \beta = \frac{AE}{DE}$

$$\begin{aligned} \Rightarrow \tan 30^\circ &= \frac{\left(\frac{3 + \sqrt{3}}{2} \right) h}{DE} \\ \Rightarrow \frac{1}{\sqrt{3}} &= \frac{\sqrt{3} \left(\frac{1 + \sqrt{3}}{2} \right) h}{DE} \\ \Rightarrow DE &= \frac{3}{2} (1 + \sqrt{3}) h \quad \dots(i) \end{aligned}$$

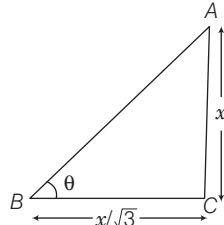
In right angled $\triangle BDE$,

$$\begin{aligned} \tan \theta &= \frac{BD}{DE} = \frac{h}{DE} \\ \Rightarrow \tan \theta &= \frac{h}{\frac{3}{2} (1 + \sqrt{3}) h} [\text{from Eq. (i)}] \\ \Rightarrow \tan \theta &= \frac{2}{3} \frac{(\sqrt{3} - 1)}{(\sqrt{3} + 1)(\sqrt{3} - 1)} \\ \Rightarrow \tan \theta &= \frac{2(\sqrt{3} - 1)}{3 \cdot 2} \end{aligned}$$

$$\Rightarrow \tan \theta = \frac{(\sqrt{3} - 1)}{3} \cdot \frac{\sqrt{3}}{\sqrt{3}}$$

$$\therefore \tan \theta = \frac{(3 - \sqrt{3})}{3\sqrt{3}}$$

34. (c) Let θ be the angle of elevation,

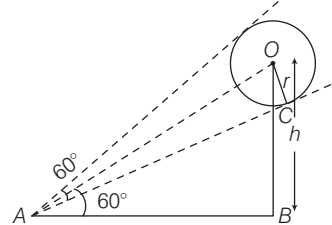


In right angled $\triangle ABC$,

$$\tan \theta = \frac{AC}{BC} = \frac{x}{x/\sqrt{3}} = \frac{\sqrt{3}x}{x} = \sqrt{3}$$

$$\begin{aligned} \text{Here, } \tan \theta &= \sqrt{3} = \tan 60^\circ \\ \therefore \theta &= 60^\circ [\because \tan 60^\circ = \sqrt{3}] \end{aligned}$$

35. (c) Given, radius of circle $(OC) = r$



In right angled $\triangle ABO$,

$$\sin 60^\circ = \frac{OB}{AO} \Rightarrow AO = \frac{OB}{\sin 60^\circ} \quad \dots(i)$$

Now, in right angled $\triangle AOC$,

$$\sin \frac{60^\circ}{2} = \frac{OC}{AO} \Rightarrow AO = \frac{OC}{\sin 30^\circ} \quad \dots(ii)$$

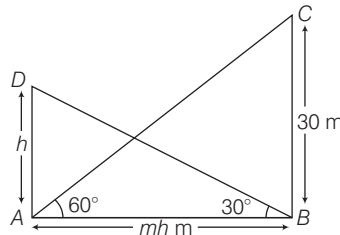
From Eqs. (i) and (ii),

$$\frac{OB}{\sin 60^\circ} = \frac{OC}{\sin 30^\circ} \Rightarrow \frac{h}{\frac{\sqrt{3}}{2}} = \frac{r}{\frac{1}{2}}$$

$$\therefore h = \sqrt{3} r$$

36. (b) Let h m be the height of shorter tower and the distance between the two towers is mh m.

Given, $\angle ABD = 30^\circ$ and $\angle BAC = 60^\circ$



In right angled $\triangle ABD$,

$$\tan 30^\circ = \frac{h}{mh} \Rightarrow \frac{1}{\sqrt{3}} = \frac{1}{m}$$

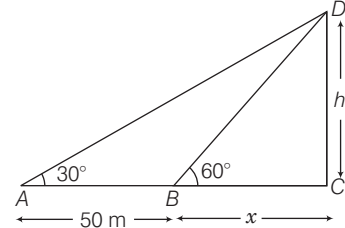
$$\Rightarrow m = \sqrt{3}$$

Hence, the value of m is $\sqrt{3}$.

37. (b) Let h m be the height of the tower and BC be x m.

In right angled $\triangle BCD$,

$$\begin{aligned} \tan 60^\circ &= \frac{DC}{BC} = \frac{h}{x} \\ \Rightarrow \sqrt{3} &= \frac{h}{x} \Rightarrow h = x\sqrt{3} \quad \dots(i) \end{aligned}$$



Now, in right angled $\triangle ACD$,

$$\begin{aligned} \tan 30^\circ &= \frac{DC}{AC} \\ &= \frac{DC}{AB + BC} = \frac{h}{50 + x} \end{aligned}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{x\sqrt{3}}{50 + x} \quad [\text{from Eq. (i)}]$$

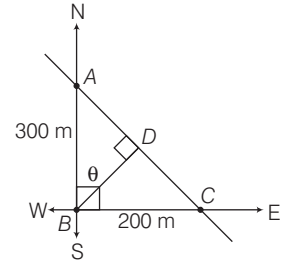
$$\Rightarrow 50 + x = 3x \Rightarrow x = 25 \text{ m}$$

$$\therefore h = 25\sqrt{3} \text{ m}$$

Hence, the height of tower is $25\sqrt{3}$ m.

38. (c) Let person be at point C and observes a tower in West direction at B .

$$\therefore BC = 200 \text{ m}$$



He walks some distance and reach at A . Now, he observe tower in South direction at B .

$$\therefore AB = 300 \text{ m}$$

Let BD be the shortest distance of tower from the road, which is a perpendicular distance.

If $\angle ABD = \theta$, then

$$\angle CBD = 90^\circ - \theta$$

[$\because$ angle between S and $W = 90^\circ$]

In right angled $\triangle ADB$,

$$\cos \theta = \frac{BD}{AB} \Rightarrow \cos \theta = \frac{BD}{300} \quad \dots(i)$$

In right angled $\triangle CDB$,

$$\begin{aligned} \cos (90^\circ - \theta) &= \frac{BD}{BC} \\ \Rightarrow \sin \theta &= \frac{BD}{200} \quad \dots(ii) \end{aligned}$$